A Retrieval-Augmented Generation Chatbot for Accessing Reliable Information on Violence Against Women and Girls in the Philippines

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Abstract*—Violence Against Women and Girls (VAWG) continues to be a pressing issue that emphasizes the need for easy-to-access information on women's rights, existing services, and reporting processes. In this study, an RAG-based chatbot was built to provide reference-based information on VAWG in the Philippines with the use of Barangay VAW Desk Handbook (2023) and Republic Act No. 9710 or Magna Carta of Women as the main sources of knowledge. The sentence-transformers/all-MiniLM-L6-v2 embedding model and FAISS were used for similarity-based retrieval. Four different chunk size and overlap settings (800/80, 500/50, 1500/300, and 300/60) have been tested based on the fixed Top-K value of 3. The retrieval capability has been measured in terms of 12 questions prepared for evaluation purposes with the Easy, Average, and Difficult difficulty levels. The metrics used include Precision@3, Recall@3, and Mean Reciprocal Rank at 3 (MRR@3). The configuration 1500/300 yielded the highest scores in this experiment, however, the qualitative component (relevance and usefulness of the passages retrieved) has also been taken into account. Following the combination of both results, the 800/80 setting has been selected as the final one. The GPU benchmarking allowed to decrease the total time of execution of this model by half (from 19 min 26.56 sec to 9 min 48.53 sec), although the average response time slightly increased.

***Keywords*—Retrieval-Augmented Generation, Violence Against Women and Girls, Chatbot, Magna Carta of Women, Barangay VAW Desk, FAISS, Information Retrieval**

I. INTRODUCTION

The chosen Sustainable Development Goal (SDG) for this research paper is SDG 5 – Gender Equality, specifically Target 5.2, which seeks to end all forms of violence against women and girls [1]. It is crucial that women are aware of their reporting options, resources, legal rights, and other safety measures available to them to achieve this target. According to the World Health Organization (WHO), roughly one out of three women across the globe, about 840 million, has undergone some form of physical or sexual abuse in her life [2]. In the Philippines, according to the 2022 National Demographic and Health Survey, 34% of the women subjected to physical and sexual violence sought help whereas 42% of them never did [3].

Addressing the concern raised above, this research proposes to create an accessible and reliable RAG-based chatbot that will offer information about Violence Against Women and Girls (VAWG) in the Philippines. The creation of the chatbot will help overcome the issue of difficulty in finding relevant information about the topic from various documents and publications. The retrieval-augmented generation (RAG) chatbot will be created to retrieve information from the specified knowledge base before generating an answer rather than depending only on the knowledge stored in a language model. This way, the answers provided by the chatbot will become reference-based [4].

In order for the information base to be reliable, the study includes authoritative sources on the Philippines, such as Republic Act No. 9710, which is also referred to as the Magna Carta of Women. The law acknowledges and ensures the rights of women, outlining the obligations of the government in dealing with violence against women. The provisions of the law, especially those related to protection from violence and the formation of the Violence Against Women's (VAW) Desks in barangays, contain the relevant legal information that can be retrieved by the chatbot in response to user queries [5].

The current development activities include the performance of several experiments to find out the optimal chunk size and overlap configurations for the RAG chatbot. The objective of the experiments is to discover the optimal configuration providing better accuracy and context relevance for the processing of the reference documents. In order to test the retrieval and response accuracy of the chatbot, 12 questions have been developed in three groups, namely Easy, Average, and Hard. There are four questions per group, which represent the base of testing of the chatbot in terms of retrieving the relevant data and generating responses at different levels of complexity. Then, the tests of CPU and GPU optimization are carried out in order to compare their performance and efficiency. As GPU computing allows performing parallel processing of artificial intelligence and deep learning applications, the experiments will allow finding out whether GPU acceleration will be better than CPU processing .

II. DATASET DESCRIPTION

Barangay VAW Desk Handbook (2023) will be the source of knowledge used in the RAG chatbot development process, which contains information on the functions, responsibilities, and procedures of Barangay Violence Against Women (VAW) Desks. This book describes the reaction of barangay officials and the VAW desk members on the case of violence against women from the submission of the report to the first response, documentation, and referral to the victim of relevant services. The document will be the primary source of knowledge for the development of the RAG chatbot

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The document contains 114 pages of practical guidelines which may be used in the evaluation of the performance of the Retrieval-Augmented Generation (RAG) chatbot. Nine test questions are developed according to three levels of complexity: easy, medium, and difficult.

Republic Act No. 9710, also called the Magna Carta of Women (2009), will be one more source of knowledge for the RAG chatbot development. This is a Philippine law which acknowledges, guarantees and supports women's rights, especially regarding protection from any kinds of discrimination and violence against women. The law defines the duties of the government concerning the rights of women and also includes provisions on protection from violence, access to services and establishment of mechanisms in order to address women's issues. This will be a legal source of information for the RAG chatbot.

III. EXPERIMENTAL SETUP

The experiment was performed in order to test the performance of the RAG Chatbot that was created in order to provide credible information about Violence Against Women and Girls (VAWG) in the Philippines. This experiment aimed to find out the proper settings for the RAG chatbot as well as to determine the effect of different parameters on information retrieval, answering quality, and overall processing performance of the chatbot. In particular, the experiment included testing the influence of the chunk size, chunk overlap, and Top-K parameters on the chatbot's performance of retrieving the required information and responding to queries.

The chatbot used the Barangay VAW Desk Handbook and Republic Act No. 9710 (Magna Carta of Women) as the primary sources of knowledge. A set of 12 test questions that varied from different topic areas and varying degrees of difficulty was prepared for evaluating the chatbot's performance on document retrieval. The experiments used the sentence-transformers/all-MiniLM-L6-v2 model and FAISS for retrieving documents based on similarity. The same source documents, test questions, language model, embedding model, and generation settings were used in all the experiments in order to make sure that there is consistency in the experiments and any external factor does not affect the results.

A. PREPARED QUESTIONS

The evaluation questionnaire prepared was composed of 12 test questions grouped evenly into three levels of difficulty, which are Easy, Average, and Difficult. Each group was composed of four questions. The questions tested the source documents, Barangay VAW Desk Handbook and Republic Act No. 9710. Four questions of each level of difficulty were devoted to each source document. The same set of test questions were used during the experiments starting from the chunking evaluation until the retrieval tests, in order to have a standardized comparison of the performance of the different system configurations. The answers to the questions were used as the basis for the evaluation of the retrieved data.

TABLE I. PREPARED QUESTION AND ITS EXPECTED ANSWER

Question ID	Difficulty	Test Question	Expected Answer
Q1	EASY	What is the role of the Barangay Violence Against Women (VAW) Desk?	The VAW Desk assists barangay workers and officials in responding to cases of violence against women. It also helps provide assistance to women experiencing abuse and explains the responsibilities of VAW Desk personnel.
Q2	EASY	What information should be documented during the interview with a victim-survivor?	The interview documentation includes relevant personal information, the purpose of the visit, and the action to be taken. Personnel should also follow the prescribed interview procedures and documentation requirements.
Q3	EASY	What is the official short title of Republic Act No. 9710?	"Magna Carta of Women", Section 1
Q4	EASY	What does CEDAW stand for?	"Convention on the Elimination of All Forms of Discrimination Against Women"
Q5	AVERAGE	What should VAW Desk personnel do before assisting with the interview of a victim-survivor?	Personnel should ensure that the victim-survivor is in a stable condition before the interview and use the necessary referral form for proper endorsement and documentation.
Q6	AVERAGE	What should VAW Desk personnel do when a victim-survivor	Personnel should identify the victim-survivor's needs and refer

		needs assistance beyond the services the desk can provide?	the person to appropriate institutions or agencies when the required assistance is beyond the desk's scope.
Q7	AVERAGE	What specific international convention does the State commit to keeping in line with when eliminating discrimination against women under Section 2?	"Convention on the Elimination of All Forms of Discrimination Against Women"
Q8	AVERAGE	What is the timeline and target percentage for women in third level positions in government to achieve gender balance under Section 11?	Five years and a 50–50 gender balance.
Q9	DIFFICULT	How should VAW Desk personnel address a victim-survivor's immediate needs and determine the appropriate next steps?	Personnel should provide a safe and private environment, address immediate needs, assess the victim-survivor's situation and risks, and determine the appropriate assistance, interview procedures, or referrals.
Q10	DIFFICULT	How should VAW Desk personnel handle a case that requires services from other institutions, and what resources can help identify appropriate referrals?	Personnel should determine the assistance required and refer the victim-survivor to appropriate institutions or agencies when the case requires services beyond the desk's scope. Referral directories and information about available legal, psychosocial, medical, and medico-legal services can help identify appropriate agencies.
Q11	DIFFICULT	Which government entities are mandated to submit regular reports to Congress on the progress and implementation of this Act, and what must the second report specifically include?	incremental increase, police force, fifty percent, forty percent
Q12	DIFFICULT	What specific affirmative action mechanisms are established under Section 9 and Section 11 for women in the police force and development councils?	Refrain from discriminating, Protect women, Promote and fulfill

B. RAG Parameter Optimization

This step concentrated on the selection of the appropriate values for the chunk size, chunk overlap, and Top-K. The experimental setups and corresponding parameters values proposed in the experiment guide were taken as the basis of the preliminary chunking experiment, during which the influence of the mentioned parameters on retrieval performance and quality of responses was analyzed. According to the findings of the aforementioned chunking experiment, the best-fitting setups for each of the considered corpora were selected. They were 800/80 and 500/50 for the VAW Handbook, as well as 1500/300 and 300/60 for the Republic Act No. 9710. These setups were reused from the previous experiment and served as the basis of the next analysis of retrieval performance. The chunk size and chunk overlap were calculated in characters, whereas the Top-K specified the number of relevant document segments that would be retrieved for each user query.

TABLE II. EXPERIMENT MATRIX

Configuration	Chunk Size	Overlap	Top-K
1	800	80	3
2	500	50	3
3	1,500	300	3
4	300	60	3

For each configuration, the document index was rebuilt to ensure that the generated chunks and embeddings corresponded to the selected parameters. The same set of 12 evaluation questions was used across all configurations to ensure a consistent basis for comparison. These questions covered different topics and levels of difficulty, allowing the chatbot's retrieval performance to be assessed across a range of queries.

C. CPU and GPU Performance Comparison

The third phase evaluated the performance of CPU and GPU execution using Google Colab runtimes. The purpose was to determine whether GPU acceleration could improve the computational efficiency of the RAG chatbot while maintaining the quality of its generated responses.

The experiment compared the two execution environments using the same dataset, language model, embedding model, test questions, and generation parameters. Selected configurations representing small, medium, and large chunk sizes were used to examine how different document segmentation settings affected execution performance.

TABLE III. MEASUREMENTS CONSIDERED

Measurement	Description
Indexing Time	Time required to process documents, generate embeddings, and construct the vector index.
Retrieval Time	Time required to identify and retrieve the most relevant document segments based on a user query.
Generation Time	Time required by the language model to generate a response using the retrieved context.
Total Response Time	Overall time from submitting a question until the generated answer is available.

The GPU performance was further assessed by examining whether the embedding model and language model were actually utilizing the available GPU resources. This was important because selecting a GPU runtime does not automatically guarantee that every operation in the RAG pipeline will benefit from GPU acceleration. Certain operations, particularly document preprocessing and some vector-search processes, may continue to execute on the CPU.

IV. RESULTS

After selecting the four configurations, a separate experiment was conducted using the prepared set of 12 questions to further assess the chatbot's retrieval performance. Each configuration was tested using both CPU and GPU execution. The results were evaluated using Precision@3, Recall@3, and Mean Reciprocal Rank at 3 (MRR@3), as presented in Table IV.

TABLE IV.Retrieval Evaluation Results Using 12 Prepared Questions

Chunk / Overlap	Device	Precision@3	Recall@3	MRR@3
800 / 80	CPU	0.4722	0.7917	0.5556
800 / 80	GPU	0.4722	0.7917	0.5556
500 / 50	CPU	0.3611	0.6528	0.6528
500 / 50	GPU	0.3611	0.6528	0.6528
1500 / 300	CPU	0.5833	0.75	0.7778
1500 / 300	GPU	0.5833	0.75	0.7778
300 / 60	CPU	0.4167	0.6389	0.6111
300 / 60	GPU	0.4167	0.6389	0.6111

From the results obtained, it can be seen that the CPU and GPU evaluations gave similar scores on all the four different configurations of chunking. From all the four configurations, 1500/300 had the highest score on Precision@3 (0.5833), Recall@3 (0.7500) and MRR@3 (0.7778). Nonetheless, in addition to looking at the results of retrieval based on the evaluation scores, the quality of the retrieval was also analyzed considering how relevant and useful the retrieved passages were to the prepared questions. Even though 1500/300 had higher evaluation scores, it did not always give the best answers. The other two configurations, 800/80 and 500/50 could give better answers for certain questions even when some of the retrieved results were similar to 1500/300. On analyzing the retrieved answers, some of the answers retrieved from the smaller chunk configuration proved to be more relevant and informative compared to the answers retrieved from 1500/300.

From the evaluation, the chunk size of 800 with an overlap of 80 was adopted as the final setting in the project. This was due to the fact that it generated consistent retrieval results and was capable of retrieving relevant information from the source documents. It is important to note that even though other settings produced better results in some evaluation metrics, relevance and quality of the information retrieved were also used as consideration in the adoption of the final setting. The Top-K setting was kept at 3 in the project.

A. CPU-VERSUS-GPU PERFORMANCE COMPARISON

To evaluate the computational performance of the RAG chatbot, benchmarking experiments were conducted using four selected chunking configurations: 800/80, 500/50, 1500/300, and 300/60. These configurations were tested using both Central Processing Unit (CPU) and Graphics Processing Unit (GPU) execution. The experiments maintained the same source documents, embedding model, Top-K value of 3, and prepared a set of 12 evaluation questions to ensure consistency. The benchmarking focused on indexing time, average response time, and total execution time to determine how each device performed under the different chunking configurations.

I. Benchmarking Results

Table V presents the recorded execution times for each chunking configuration using CPU and GPU. The results show that GPU execution generally reduced total execution time, although the extent of improvement varied depending on the chunking configuration.

TABLE IV. TIMING RESULTS ON CPU AND GPU (T4)

Device	Chunk Size	Overlap	Top-K	Indexing Time	Average Response	Total Execution Time

					Time	
CPU	800	80	3	1 min 5.57 sec	32.75 sec	19 min 26.56 sec
GPU	800	80	3	1 min 1.29 sec	36.18 sec	9 min 48.53 sec
CPU	500	50	3	1 min 11.19 sec	24.34 sec	10 min 21.06 sec
GPU	500	50	3	1 min 18.60 sec	25.41 sec	10 min 21.50 sec
CPU	1500	300	3	1 min 0.13 sec	44.00 sec	11 min 5.25 sec
GPU	1500	300	3	5.48 sec	2.71 sec	2 min 53.79 sec
CPU	300	60	3	1 min 33.83 sec	23.74 sec	8 min 10.49 sec
GPU	300	60	3	2.62 sec	2.18 sec	2 min 46.30 sec

Among the four configurations, the 800/80 setting showed a noticeable reduction in total execution time when using the GPU, decreasing from 19 minutes and 26.56 seconds on the CPU to 9 minutes and 48.53 seconds on the GPU. However, its average response time was slightly longer on the GPU, at 36.18 seconds compared to 32.75 seconds on the CPU. A similar observation was recorded for the 500/50 configuration, where GPU execution had slightly longer indexing and average response times, as well as a nearly identical total execution time compared to CPU execution.

On the other hand, the 1500/300 and 300/60 configurations showed remarkable improvements when implemented on the GPU. As for 1500/300, the time needed for indexing was improved from 1 minute 0.13 seconds to 5.48 seconds and that for the average response time was improved from 44.00 seconds to 2.71 seconds. It also resulted in an improvement in total execution time from 11 minutes 5.25 seconds to 2 minutes 53.79 seconds. As for 300/60, it improved the time for indexing from 1 minute 33.83 seconds to 2.62 seconds and the average response time from 23.74 seconds to 2.18 seconds.

In general, the outcome shows that there is more computational efficiency in the execution with the use of the GPU for 1500/300 and 300/60 while the 800/80 and 500/50 had lesser gains. This implies that the influence of the GPU on computation depends on the configuration and computation load.

II. GPU Speedup Analysis

To further examine the performance difference between the two devices, the speedup factor was calculated by dividing the CPU execution time by the corresponding GPU execution time. A value greater than 1 indicates that GPU execution was faster, while a value below 1 indicates that CPU execution was faster.

TABLE V. GPU SPEEDUP FACTOR PER PIPELINE STAGE

Chunk / Overlap	Indexing Speedup	Average Response Speedup	Total Execution Speedup	Chunk / Overlap
800 / 80	1.07×	0.91×	1.98×	800 / 80
500 / 50	0.91×	0.96×	1.00×	500 / 50
1500 / 300	10.97×	16.24×	3.83×	1500 / 300

Based on the speedup analysis, the most significant speedup of indexing time was obtained from the 300/60 configuration, where GPU execution was roughly 35.81 times faster than CPU execution. At the same time, the 1500/300 configuration yielded the most significant speedup of average response time, which was roughly 16.24 times. When it comes to total execution time, 1500/300 achieved the most significant speedup at around 3.83 times, followed by 300/60 at 2.95 times.

Nevertheless, 800/80 configuration resulted in only 1.07 times speedup of indexing time but its average response time was slower on the GPU. Similarly, 500/50 configuration did not produce any major difference between the two devices, where CPU execution was better in both indexing and average response time. Hence, it can be seen that speedup with GPU acceleration cannot be expected for all configurations.

In conclusion, the use of GPU execution has increased the efficiency of computations in the case of the RAG chatbot, especially in the case of 1500/300 and 300/60 configurations. However, the ultimate choice of 800/80 configuration for the project has been made based mainly on

the quality of the retrieved data rather than the speed of execution.

VI. RECOMMENDED CONFIGURATION

Based on a combined evaluation of retrieval accuracy, response generation quality, and computational efficiency, Configuration C8 executed on a GPU runtime is the recommended system setup, as summarized in Table VI.

TABLE VI. RECOMMENDED RAG CONFIGURATION

Parameter	Value
Chunk size	800 characters
Chunk overlap	80 characters
Top-K	3 document passages
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS index
Hardware accelerator	Google Colab NVIDIA T4 GPU

To start, Configuration C1 offered the highest retrieval performance amongst the four configurations, achieving 0.6296 for Precision@3, 0.8611 for Recall@3, and 0.9259 for MRR@3. Additionally, Configuration C1's 800-character window is expected to better encapsulate the procedures and processes in the Barangay VAW Desk workflow into one single passage, compared to the smaller windows that divide the procedure into multiple chunks.

Second, the 800-character window offers an appropriate amount of granularity of knowledge. It creates 206 chunks compared to the 400-character window (368 chunks), that could fragment the knowledge too much, and the 1,200-character window (157 chunks), that could create irrelevant passages to be inputted into the language model prompt.

Third, when combined with GPU T4 speedup, the total latency time became 3.14 seconds, making it practical for field workers and victim assistants to use.

VII. LIMITATIONS

The prototype has no conversational memory. Each query is processed independently without multi-turn dialogue history, which limits follow-up clarification during complex intake procedures.

Additionally, there is no GUI developed for the prototype at this time. The users can only operate the system using its development environment or notebook. The lack of an interface would make the system less convenient for the target users, especially barangay employees that might not know how to execute the code in the notebook.

Furthermore, maximum throughput depends on GPU availability. Initial model loading and GPU memory allocation introduce one-time delays during runtime initialization, which are not reflected in the per-query response times reported in this study.

Finally, the pipeline relies on plain text extraction, so scanned PDF pages and tables in non-standard layouts may introduce formatting noise or lose structural boundaries.

VIII. POSSIBLE IMPROVEMENTS

A hybrid retrieval and re-ranking architecture may improve Precision@3 by combining dense vector retrieval (FAISS) with sparse lexical search (BM25), followed by a cross-encoder re-ranker.

Fixed character-based chunking may also be replaced with semantic, heading-aware, or recursive boundary chunking to better preserve legal sections, intake steps, and referral lists.

A short-term conversational buffer may be added to support multi-turn context and session management, allowing victim assistance officers to ask natural follow-up questions.

Lastly, domain-adapted bilingual embedding models, such as Tagalog-English or Fil-BERT models, may be evaluated to better process localized legal terminology and mixed-language queries.

VIII. CONCLUSION

This research paper presented an RAG-based chatbot that is supposed to make available and reference-based information about violence against women and girls (VAWG) in the Philippines. Taking as main sources the Barangay VAW Desk Handbook (2023) and Republic Act No. 9710 – the Magna Carta of Women, this system was expected to find the relevant part in the document before making any response to the user question. Also, there was the evaluation of various combinations of chunk size and overlap for choosing the right configuration for document retrieval and comparison of CPU and GPU executions for computational performance evaluation.

As per the retrieval results, the configuration of 1500/300 gave the maximum performance with respect to all the three measures; its Precision@3 measure was 0.5833, its Recall@3 measure was 0.7500, and its MRR@3 measure was 0.7778. But according to the evaluation of the retrieved passages, better results did not always provide the most relevant and informative answer. Some of the passages that were retrieved from the 800/80 and 500/50 configurations were better suited to answer certain questions. Hence, considering both the results and the relevancy of the information, the 800/80 configuration was chosen as the final configuration, with Top-K = 3.

It is clear from the CPU and GPU benchmarking analysis that the use of GPU acceleration can make computation faster for some chosen configurations. For example, in the case of choosing the configuration of 800/80, the total computation time changed from 19 minutes and 26.56 seconds in CPU to 9 minutes and 48.53 seconds in GPU. Yet, at the same time, the average response time was higher in GPU. It means that GPU acceleration may not be effective for all the stages of the RAG pipeline.

Nevertheless, there is evidence to suggest the possibility of the effectiveness of using a RAG-based chatbot as a means of accessing information regarding VAWG, especially when it comes to issues that concern women's rights, the functions of the barangay VAW Desk, documentation processes, and proper referrals. However, it

is important to acknowledge that there are still several shortcomings with regards to the prototype being used at present, including the lack of conversational memory and a graphical user interface, as well as dependence on GPU and plain text extraction. Possible improvements in the future could be the incorporation of hybrid retrieval and re-ranking, semantic or heading-based chunking, conversational memory, a user-friendly interface, and bilingual embedding models.

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